



Fast-track Your Wireless Sensing Application Development Using MCXW71 and Low-Power Pressure Sensors

Hands-on Lab#3 Guide

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Hands-On Lab#3 Overview:

Tamper detection using low-power pressure sensor:
Securing Your Assets with MCXW71 Wireless MCU

Brief Description

This example uses **FRDM-MCXW71** and **MPL3115** pressure sensor (using sensor expansion board) to demonstrate autonomous detection of tampering/theft/abuse on device using low-power wakeup on change in pressure and transmit ALERT message via BLE wireless UART.

Details

The core idea is to leverage advanced sensor technology to protect or secure assets. For example, in industrial environment for pressure sensitive seals on containers, valves and control panels. In food, pharmaceutical and chemical industries for pressure sensitive labels. In home, building environment for pressure tampering on water/gas meters, HVACs etc.

Solution

This example demonstrates ease-of-enablement using NXP's FRDM-MCXW71 and sensors development ecosystem to accelerate prototyping for your multiple such applications.

The **Lab#3** solution comprises of

- **MCX W71** for transmitting tamper detection alerts via BLE.
- Absolute pressure, altimeter sensor **MPL3115A2** for detecting change in pressure/altitude.



Recommended Sensors

MPL3115A2S: Absolute Pressure/Altimeter Sensor

The **MPL3115A2s** is a compact, **piezoresistive, absolute pressure sensor** with an I2C digital interface. MPL3115A2 has a **wide operating range of 20 kPa to 110 kPa**. It has multiple user programmable modes such as power saving, interrupt, and autonomous data acquisition modes, including programmed acquisition cycle timing, and poll-only modes.

Features:

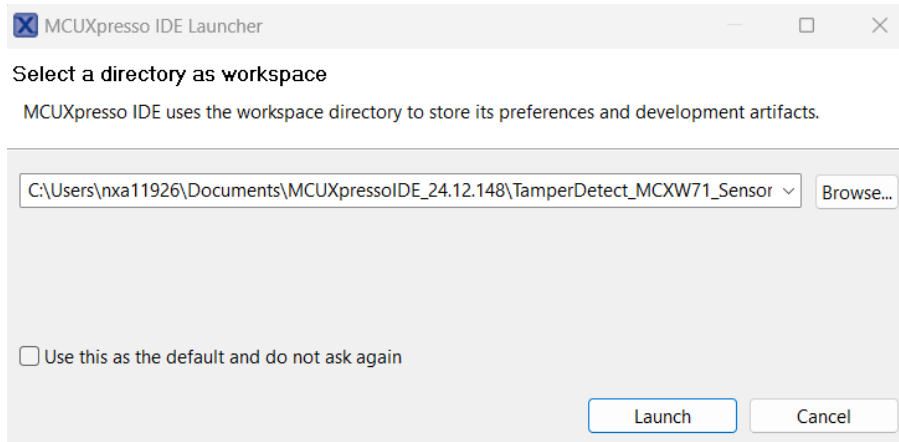
- **Pressure resolution: 1.5 Pa.**
- **Guaranteed drift over time (<20 Pa) for use in “pseudo differential mode”**
- Pressure ranges: **20 – 110 kPa** and **50–150 kPa**.
- **1.95 V to 3.6 V** supply voltage
- I²C digital interface
- Interrupt driven events.
- 32-Sample FIFO
- **Low power down to 8μA**

Target Applications:

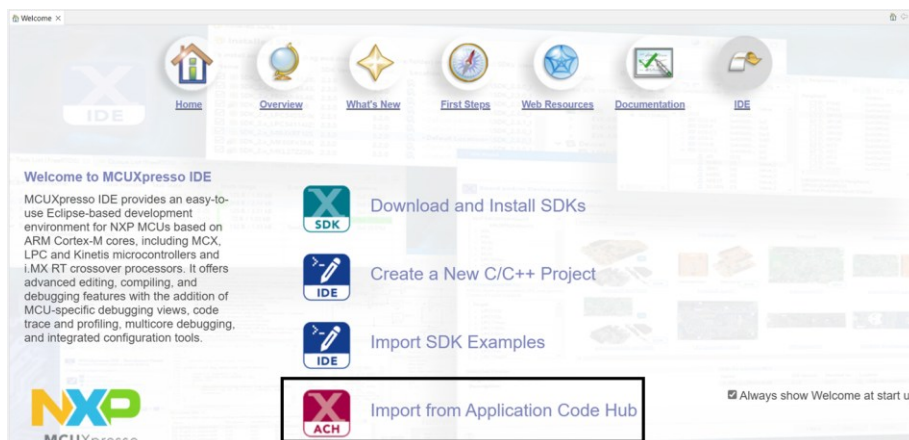
- Altimeters/navigation systems
- Weather station equipment
- Health/Activity Monitors
- Smart Inhalers
- Oxygen Concentrators
- Ventilation system

Get Hands-on Workshop Lab Examples from ACH

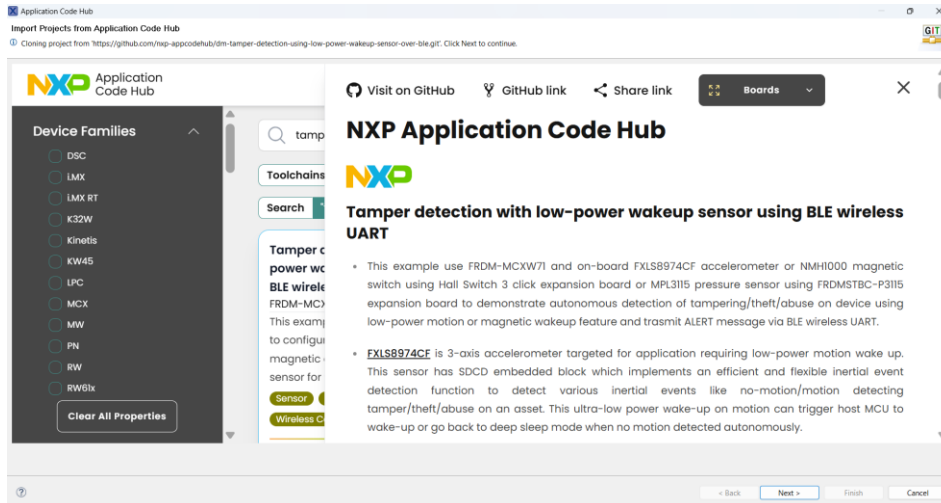
- 1) Launch MCUXpresso IDE and create workspace name as “TamperDetect_MCXW71_Sensors”.



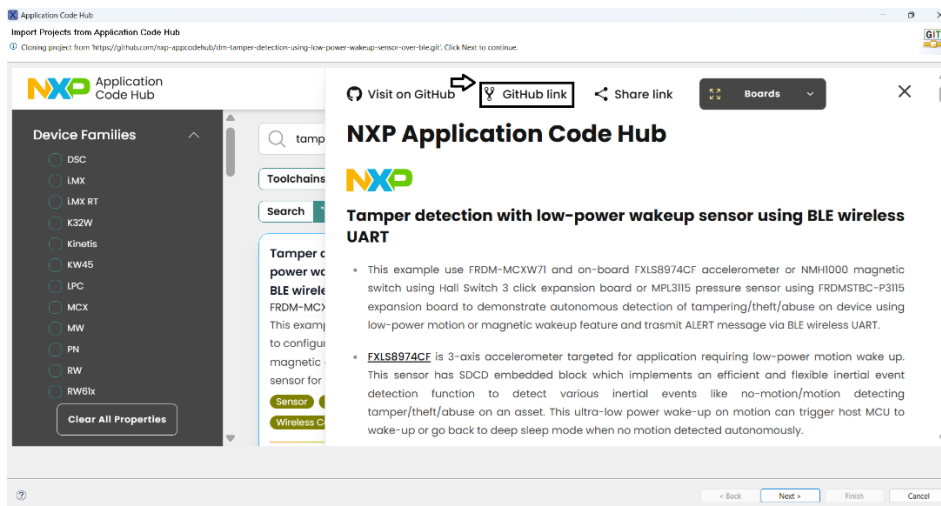
- 2) Select “Import from Application Code Hub” on the welcome screen.



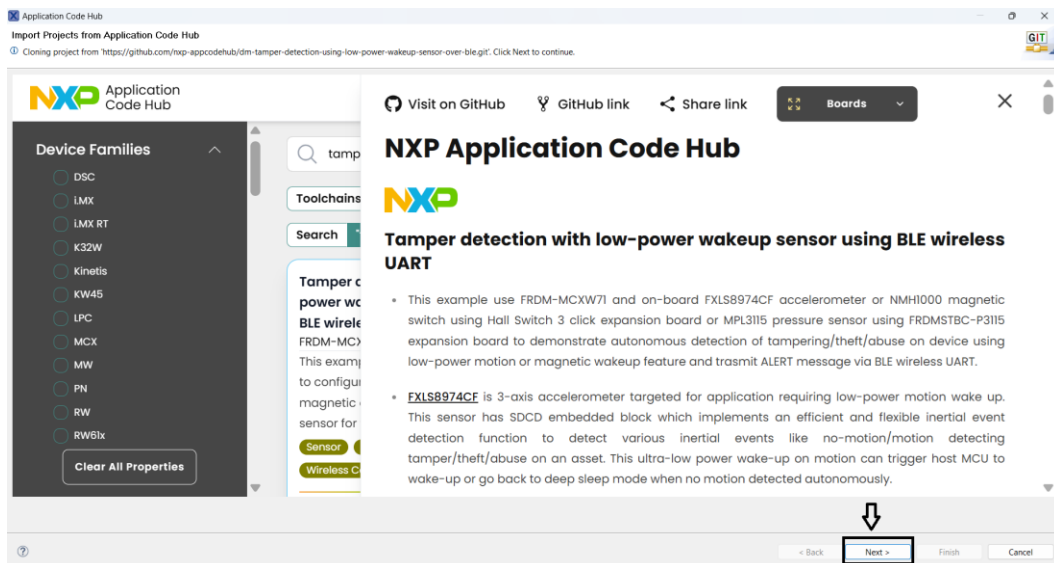
- 3) It will open a view, type in “Tamper detection FRDM-MCXW” in the search window. Select the “**Tamper Detection with low-power sensors using BLE wireless UART**” example as shown in the snapshot.



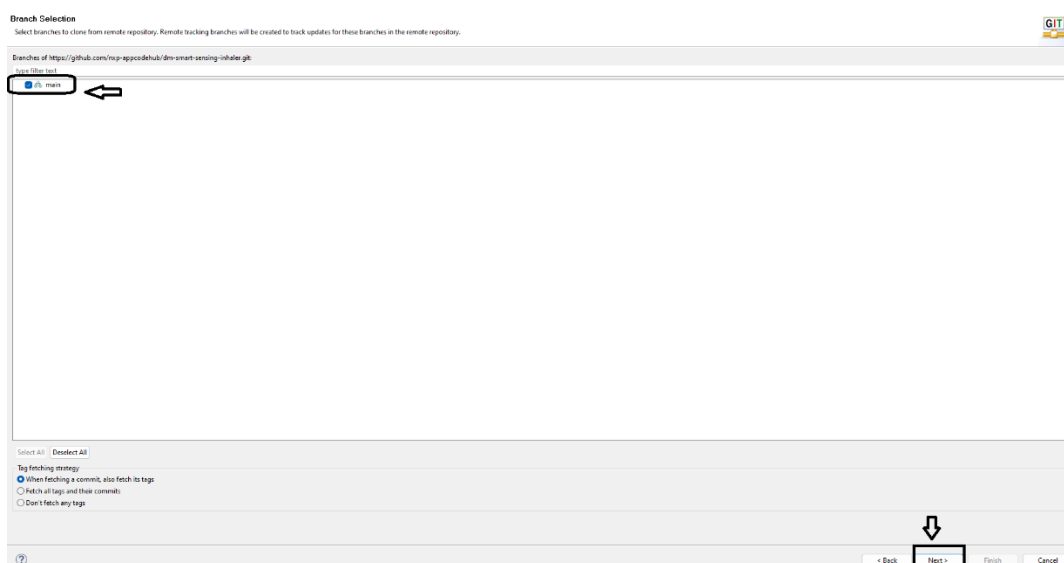
- 4) Click on “GitHub link”, it will start cloning the chosen example project into workspace folder.



- 5) The chosen example project is now getting imported into MCUXpresso IDE workspace. Click “Next”



6) Select git branch as “main” and click “Next”.



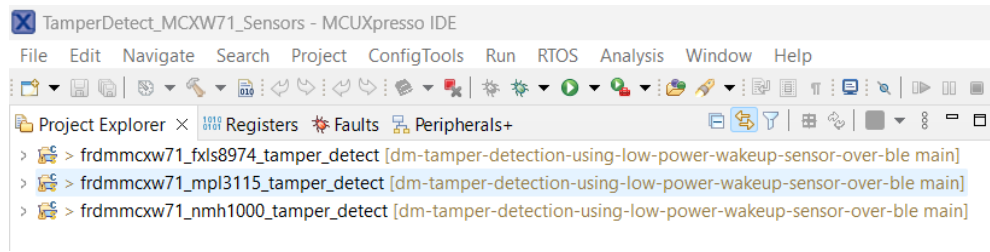
7) Select the local storage locate for cloned project. By default, the tool will select C:\Users\<userid>\git\ folder. Click “Next”.



8) Select projects available under “C:\Users\<userid>\git\dm-tamper-detection-using-low-power-wakeup-sensor-over-ble\tamper_detection_demo”. Click “Next”.

Note: Ignore any “Project SDK management” warning showing older version of SDK. Click on ok button (for all the 3 examples) and continue.

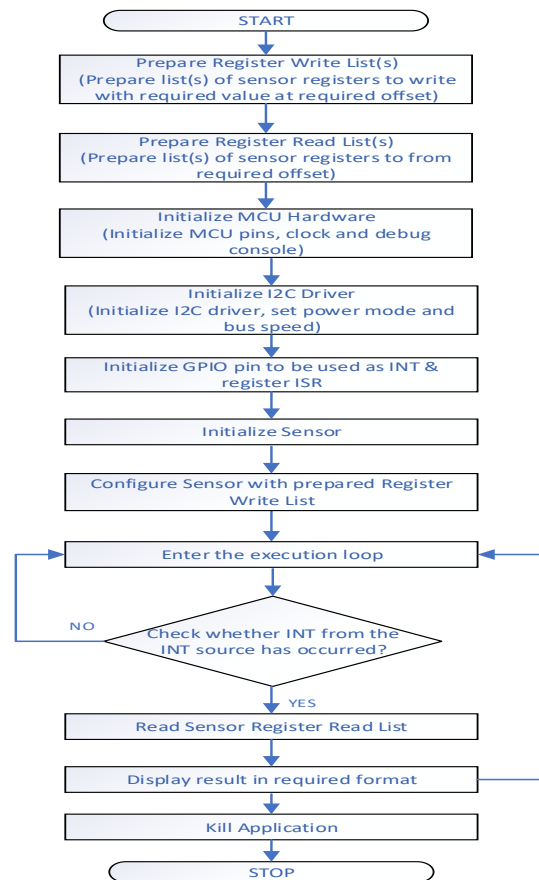
9) Click “Next”. MCUXpresso IDE will open the chosen projects.



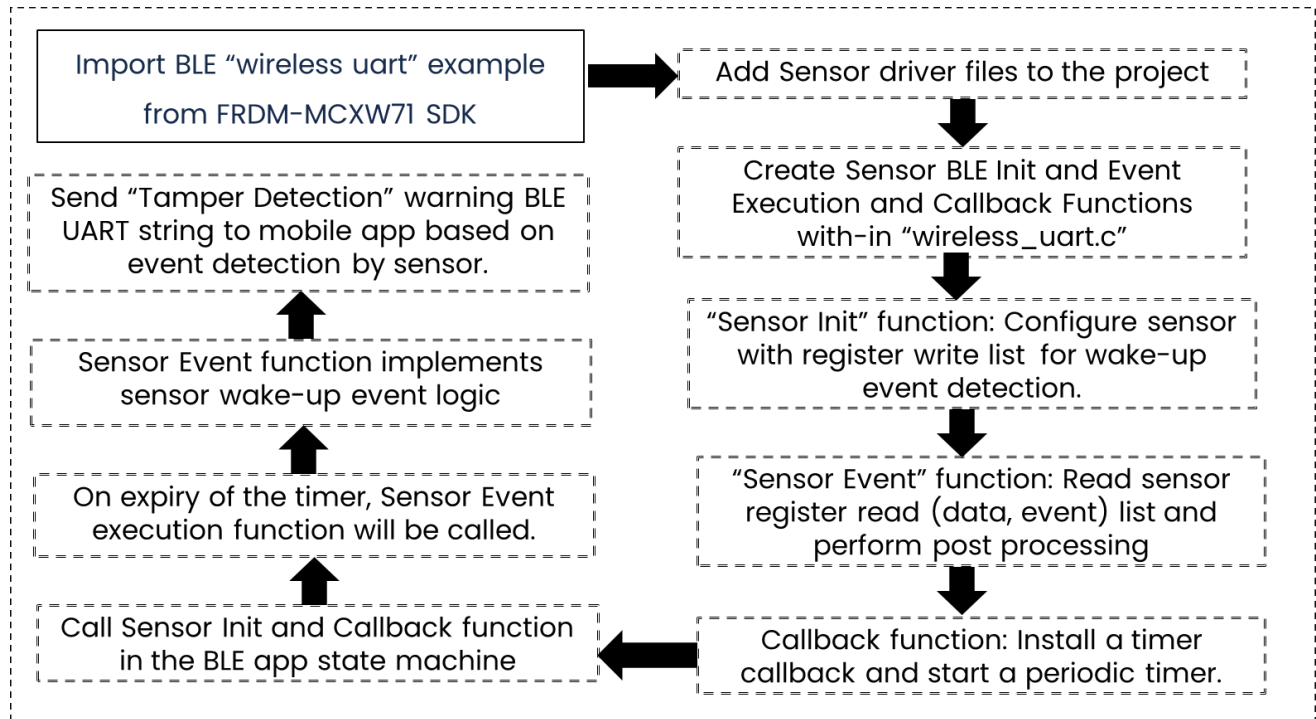
Code Walkthrough

Typical ISSDK example code flow:

- 1) Prepare register write list.
- 2) Prepare register read list.
- 3) Initialize MCU Hardware (Pins, Clock etc.)
- 4) Initialize COMM (I2C, SPI) driver
- 5) Initialize GPIO pin to be user as INT and register ISR.
- 6) Initialize Sensor, read WHOAMI
- 7) Configure Sensor with prepared register write list
- 8) Enter execution loop:
 - i. Check whether interrupt source has occurred.
 - ii. If yes, then read register read list
 - iii. Display result or upload into TSA table
 - iv. Execute post processing or other functions.



Learn how to convert SDK wireless example to wireless sensing example:



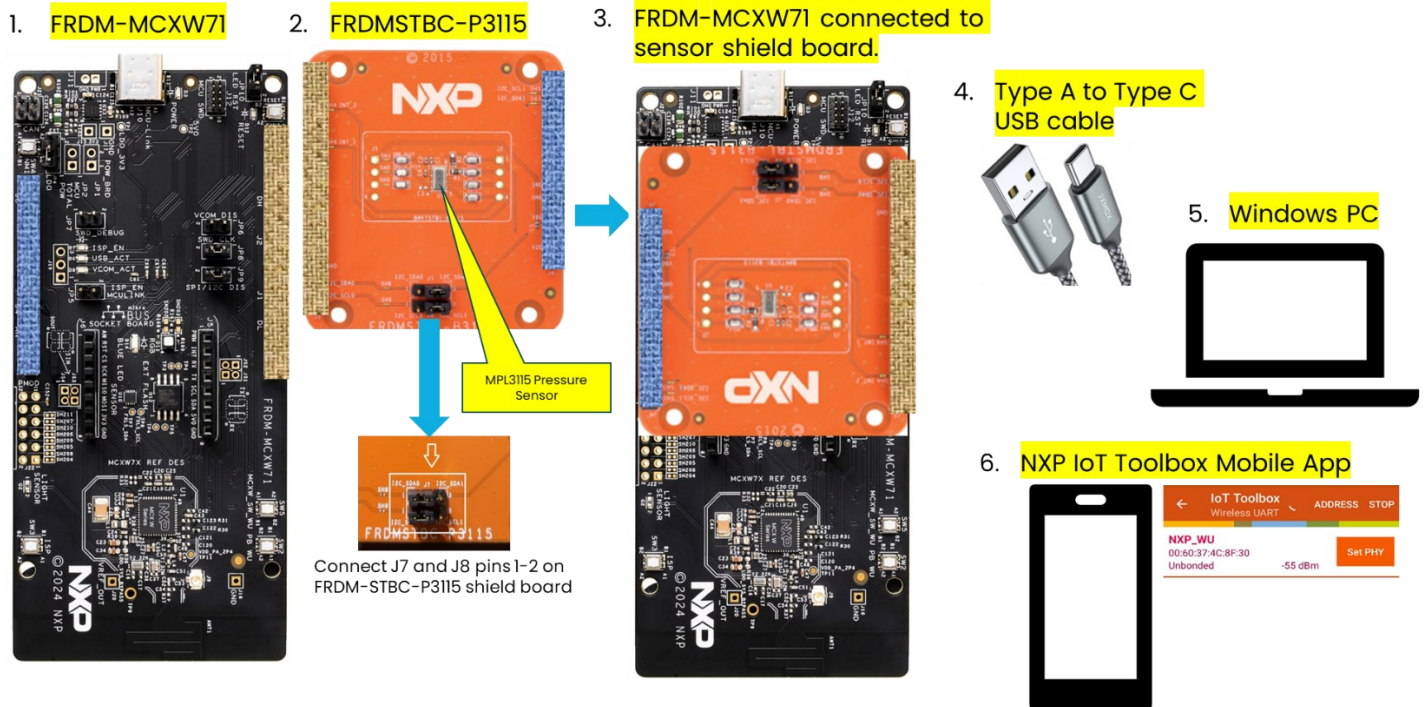


Hands-on Lab#3: Using Pressure Sensor

Compile Project:

DO IT YOURSELF

Connect HW:

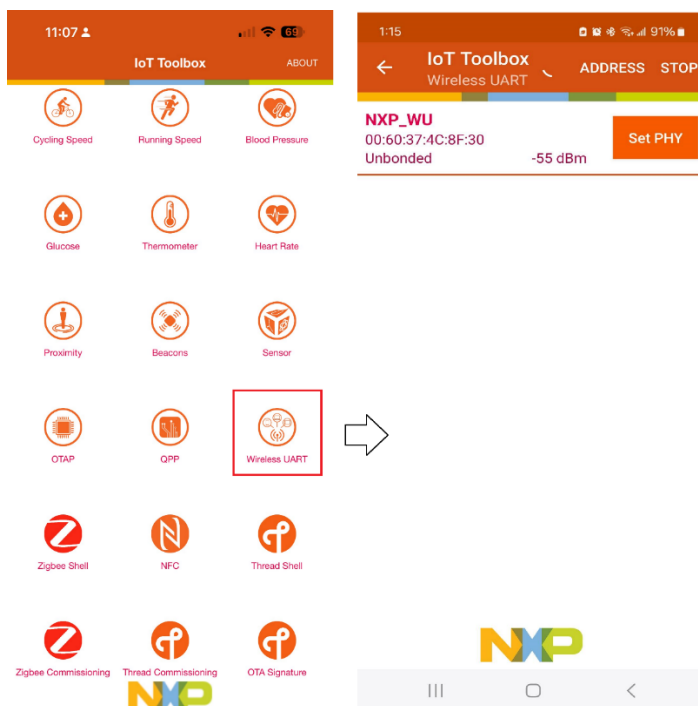


Program the target Board:

DO IT YOURSELF

Run NXP IoT Toolbox App on your mobile:

- 1) Open the installed NXP IoT Toolbox (available from Play Store and App Store) on your cell phone. Make sure Bluetooth connection is ON.
- 2) Inside the application, click on the Wireless UART widget. Scan should start automatically.
- 3) When a device called NXP_P0 or <your_id> appears, click on it. Your phone should now be connecting to the board.

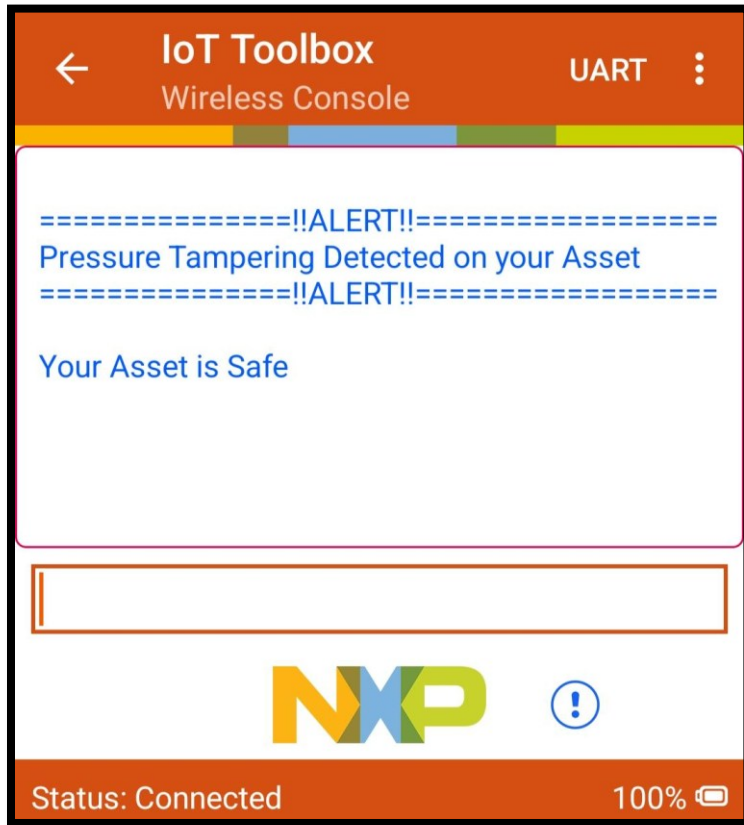


- 4) After connecting with the FRDM-MCXW71 device, the serial UART terminal will show this message:

```
RealTerm: Serial Capture Program 2.0.0.57

Wireless UART starting as GAP Peripheral.
Advertising...
Connected to device 0 as peripheral.
```

- 5) When you press/apply pressure using your index finger to the MPL3115 sensor (available on FRDMSTBC-P3115 shield board connected to FRDM-MCXW71) to show pressure tampering, MPL3115 sensor detects change in pressure and wake-up
- 6) At that point, you will see Wireless UART app showing ALERT message as shown below:



- 7) FRDM-MCXW71 board will also show "RED" LED status. The "RED" LED status will continuously remain ON till pressure tampering is detected.
- 8) After the app status shows "Your Asset is Safe", you will see "RED" LED turned OFF, at this point try putting pressure again to detect tampering.

Congratulation on successfully completing Hands-On Lab3



Congratulations!
You successfully
completed the
Hands-On Lab#3